Vancouver, BC | 604-789-1216 | navya.gadhiraju@outlook.com| Portfolio Website | LinkedIn

NAVYA GADHIRAJU

EDUCATION

University of Victoria, Victoria, BC (2024 to 2029 - Year 3)
Bachelor of Engineering in Biomedical/Mechanical Engineering (BEng)

WORK HISTORY

RESEARCH EXPERIENCE INTERN 04/2026 to 08/2026
BC Cancer (Deeley Research Centre - Fonseca Lab), Victoria, Canada

- Analyzed provincial metadata across ~9,000 bio banked samples using R, identifying a 3-day collection-to-processing median gap that initiated blood tube stability trials.
- Evaluated cfDNA concentration trends across 2,641 unique patients and tracked serial sample fluctuations across 51 Renal Cell Carcinoma patients using custom R scripts.
- Developed a visual bench protocol for the QIAGEN RNeasy FFPE Kit, now integrated into standard lab operations for paraffin-embedded tissue RNA extraction.
- Processed clinical blood samples for the VHL Biobank in a biosafety cabinet via two-stage centrifugation to separate cfDNA plasma and buffy coat fractions.

BIOMEDICAL SYSTEMS ENGINEERING INTERN 01/2026 to 04/2026
Victoria Hand Project, Victoria, Canada

- Assembled, cabled, and quality-tested 40+ 3D-printed prosthetic hands (VC300, VO300, VS300) for delivery to partner healthcare clinics in 11 countries. Led and trained 50+ volunteers.
- Designed and prototyped assistive devices compatible with different prosthetic hand models, including modifications to a mouse-clicker assistive device for desktop and laptop device accessibility
- Utilized 3D printing, 3D scanning, and CAD modelling on Autodesk Fusion 360 to support prosthetic design iteration and component improvement
- Conducted tensile testing on 3d-printed 12 TPU specimens across print orientations to evaluate yield strength and Young's modulus, guiding component material selection.
- Repaired and maintained Prusa and Ultimaker 3D printers, updating assembly documentation

RESEARCH ASSISTANT 01/2026 to 04/2026
Willerth Lab, Victoria, Canada

- Co-authored a standardized lab protocol based on a scientific manuscript on 3D bio printed neural tissue models for Parkinson's disease, adapting procedures into the Sharma Current Protocols format.
- Documented step-by-step methodologies for patient-derived hiPSC culture, bioink formulation, Aspect Biosystems RX1 3D bioprinting, and immunochemistry
- Collaborated with researchers to iterate and refine tissue engineering protocols, supporting consistent bio fabrication documentation and reproducibility. Currently being reviewed for publication

ENGINEERING PROJECTS

MRI Simulator Project 09/2025 to 12/2025

- Developed a pediatric MRI simulation project with a focus on hospital-grade materials and child-centered healthcare applications

Fortis BC x WEST Biogas Compressor design project 01/2025 to 04/2025

- Evaluated energy performance through modelling and real-world testing for an industrial biogas compressor system
- Presented project findings through an industry-facing project showcase

MedaSort 01/2025 to 04/2025

- Developed a prototype incorporating Raspberry Pi Pico, circuit analysis, and Python programming

Introba Passive Building Prototype Project with UVic WEST 09/2025 to 12/2025

- Designed a prototype structure incorporating climate-adapted passive building strategies using CAD

SKILLS

- CAD; SolidWorks, Fusion 360
- Engineering Design/principles, Prototyping, and Manufacturing
- 3D printing and scanning
- Bioinformatics
- Wet lab/equipment technique
- C++, C, R, MATLAB, and Python
- Clinical research and reporting
- Experimentation and testing
- Technical documentation
- Agile methodologies
- Cross-disciplinary communication

CERTIFICATIONS AND AWARDS

- WHIMIS
- TCPS 2: CORE 2022
- Biosafety Training; BC Cancer
- PHSA Chemical Safety Training
- Radiation, X-ray, and Laser Safety
- Blood-borne Pathogens
- Cora Arenas and Carol Artemiw Women in Engineering and CSC Award ($10,000)
- Women in Engineering and Computer Science Entrance Scholarship
- Ethics Bowl National Finalist

ACTIVITIES AND INTERESTS

- Formula One racing, UVic BME design team, woodworking, teaching skiing